PHY202 Exam 1 on chapter(s): 16, 17

Name:__________________________________

These questions assess your understanding of the material from Chapter(s) 16, 17 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. Calculate the sound intensity level in decibels for a sound wave traveling in air at 5.7°C with a pressure amplitude of 1.57 Pa. Recall that the density of air is 1.29 $\frac{kg}{m^3}$ and the speed of sound in air at 0°C is 331 $\frac{m}{s}$.

ANSWER: 94.56dB ✓

$\beta(dB) = 10\log_{10}\left(\frac{I}{I_0}\right)$

$I = \frac{(\Delta p)^2}{2\rho v_w} =$

$\frac{(1.57)^2}{2(1.29)(334.4)} = 2.86\times10^{-3}\ \frac{W}{m^2}$

$v_w = 331 m/s\sqrt{\frac{278.7}{273}} = 334.44$

$10\log_{10}\left(\frac{2.86\times10^{-3}}{1\times10^{-12}}\right)$

$= 94.56$

2. If the shock absorbers in a car go bad, then the car will oscillate at the smallest provocation, such as when going over bumps in the road and after stopping. Calculate the frequency of oscillations for a car with shoddy shock absorbers if the car's total mass is 823. kg and the force constant of the suspension system is $2.78\times10^4\ \frac{N}{m}$.

ANSWER: 0.93Hz ✓

$m = 823kg$

$k = 2.78\times10^4\ N/m$

$T_{SHO} = 2\pi\sqrt{\frac{m}{k}}$

$f = \frac{1}{T}$

$2\pi\sqrt{\frac{823}{2.78\times10^4}} = 1.081$

$f = \frac{1}{1.081} = 0.93Hz$

3. How much energy is stored in the spring of a toy gun that has a force constant of 57.0 $\frac{N}{m}$ and is compressed 13. cm?

ANSWER: 0.48J ✓

$PE = \frac{1}{2}kx^2$

$k = 57.0 N/m$

$x = 13cm$ or $0.13m$

$\frac{1}{2}(57)(0.13)^2 = 0.48J$

+3

4. Suppose you have a tube that is closed at one end. Its length is 13. cm and the air temperature is 9.1°C. What is the frequency of its first overtone?

ANSWER: 1.9×10^3 Hz

$f_n = n \frac{v_w}{4L}$ $n = 3$

$v_w = (331 m/s)\sqrt{\frac{282.1}{273}} = 336.47$

$(3)\frac{336.47}{4(0.13)} = 1941.17 Hz$

1.9×10^3

L = 13cm or 0.13m

5. When Samantha gets in her car, it lowers by 0.60 cm. If Samantha's mass is 113. kg, then what is the force constant of her car's suspension system?

ANSWER: 1.85×10^5 N/m

$F = -k\Delta x$

$\Delta x = 0.60cm$
$= 0.006m$

$F = ma$

$m = 113kg$ $a = -9.8$ kg

$ma = -k\Delta x$

$(113)(-9.8) = -k(0.006)$

$k = 184755$ N/m

$k = 1.85 \times 10^5$ N/m

5.4s one way

6. A sonar echo returns to a submarine 10.8 s after being emitted from the submarine. What is the distance from the submarine to the object, given that the speed of sound in this water is 1.52 $\frac{km}{s}$?

ANSWER: 8.21 km

$v = \frac{d}{t}$

$(5.4s)(1.52 km/s) = 8.21$

$vt = d$

+3

7. A medical imaging device produces ultrasound by oscillating with a period of 0.142 μs. What is the frequency of this oscillation?

ANSWER: 7.04×10^6 Hz ✓

$f = \frac{1}{T}$

$f = \frac{1}{0.142 \times 10^{-6}\,s}$

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7.04×10^6 Hz

$T = \frac{1}{f}$

0.142 μs

0.142×10^{-6} s

8. Ultrasound is transmitted from fat to muscle during a particular medical procedure. Calculate the percentage of the ultrasound's intensity that is reflected when the ultrasound travels from fat with a density of 828. $\frac{kg}{m^3}$ to muscle tissue with a density of 1.149×10^3 $\frac{kg}{m^3}$. The speed of ultrasound in the fat is 1.43×10^3 $\frac{m}{s}$ and the speed of ultrasound in the muscle tissue is 1.54×10^3 $\frac{m}{s}$.

ANSWER: 4% ✓

$Z = \rho v$ $\quad a = \frac{(Z_2 - Z_1)^2}{(Z_1 + Z_2)^2}$

$Z_1 = (828)(1.43 \times 10^3) = 1.18 \times 10^6$

$Z_2 = (1.149 \times 10^3)(1.54 \times 10^3) = 1.77 \times 10^6$

$a = \frac{(1.77 \times 10^6 - 1.18 \times 10^6)^2}{(1.18 \times 10^6 + 1.77 \times 10^6)^2} = 0.04$

$0.04 \times 100 = 4\%$

9. Suppose a 0.303-kg object is connected to a horizontal spring. There is friction between the object and the surface, and the coefficient of kinetic friction μ_k is 0.25. What total $\boxed{\text{distance}}$ does the object travel if it is released from rest at a distance of 36. cm from its equilibrium position? The force constant of the spring is 67.7 $\frac{N}{m}$.

ANSWER: 5.9 m ✓

$E = f \cdot d$

$\frac{E}{f} = d$

μ_k N mg

$f_k = \mu_k N$

$= \mu_k mg$

$f_k = (0.25)(0.303 kg)(9.81)$

$f_k = 0.743$

$\frac{4.39 J}{0.743} = 5.9 m$

$PE = \frac{1}{2}kx^2$

$\frac{1}{2}(67.7)(0.36)^2 = 4.39$

$36 cm = 0.36 m$

+3

10. Suppose a train sounding its 623.-Hz horn is moving at 51.9 $\frac{m}{s}$. What frequency does a person standing by the tracks hear as the train approaches? Given this particular day's temperature, the speed of sound is 321. $\frac{m}{s}$.

$f_{obs} = f_s\left(\frac{V_w}{V_w - V_s}\right)$

ANSWER: 743.15 Hz

$(623\,Hz)\left(\frac{321\,m/s}{321-51.9}\right) = 743.15$

11. Suppose you have a tube that is closed at one end. If you want its fundamental frequency to be 261. Hz and the air temperature is 0.14°C, then what should the length of the tube be?

$f_n = n\frac{V_w}{4L}$

ANSWER: 0.32m

$4Lf_n = nV_w$

$L = \frac{nV_w}{4f_n}$ $\quad L = \frac{(1)(331.08)}{4(261)} = 0.32m$

$V_w = (331)\left(\sqrt{\frac{273.14}{273}}\right) = 331.08$

12. Suppose that a 1.51×10^3 -kg car has a suspension system with a force constant $7.80 \times 10^4\ \frac{N}{m}$. If the car hits a bump and bounces with an amplitude of 11. cm, then what is its maximum vertical velocity assuming no damping occurs?

ANSWER: 0.79 m/s

$v(t) = \sqrt{\frac{k}{m}}\, x \sin\frac{2\pi t}{T}$

$v(t) = (7.187)(0.11)\sin 7.18t$

$\sqrt{\frac{k}{m}} = 7.187$

$v(t) = 0.79$

$x = 0.11m$

$m = 151 \times 10^3\,kg$

$k = 7.8 \times 10^4$

$T = 2\pi\sqrt{\frac{m}{k}}$

$= 0.874$

+3 $\quad \frac{2\pi}{0.874} = 7.18$

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1. $\beta = 94.6\ \text{dB} \because v_w = \left(331\,\frac{\text{m}}{\text{s}}\right)\sqrt{\frac{T}{273\ \text{K}}},\ I = \frac{(\Delta p)^2}{2\rho v_w},\ \beta\ (\text{dB}) = 10\log_{10}\left(\frac{I}{I_0}\right),\ I_0 = 1\times10^{-12}\,\frac{\text{W}}{\text{m}^2}$
(Chapter 17, Section (3) Sound Intensity and Sound Level - cp1e_ch17_ex2)

2. $f = 0.925\ \text{Hz} \because T_{SHO} = 2\pi\sqrt{\frac{m}{k}},\ f = \frac{1}{T}$
(Chapter 16, Section (3) Simple Harmonic Motion - cp1e_ch16_ex4)

3. $PE = 0.482\ \text{J} \because PE = \frac{1}{2}kx^2$
(Chapter 16, Section (1) Hooke's Law - cp1e_ch16_ex2)

4. $f = 1.94\times10^3\ \text{Hz} \because f_n = n\frac{v_w}{4L},\ v_w = \left(331\,\frac{\text{m}}{\text{s}}\right)\sqrt{\frac{T}{273\ \text{K}}}$
(Chapter 17, Section (5) Sound Interference and Resonance: Standing Waves in Air Columns - cp1e_ch17_ex5b)

5. $k = 1.85\times10^5\,\frac{\text{N}}{\text{m}} \because \vec{F} = -k\Delta\vec{x}.\ k = -\frac{\vec{F}}{\Delta\vec{x}}$
(Chapter 16, Section (1) Hooke's Law - cp1e_ch16_ex1)

6. $d = 8.21\times10^3\ \text{m} \because \vec{v}_{avg} = \frac{\Delta\vec{x}}{\Delta t}$
(Chapter 17, Section (2) Speed of Sound, Frequency, Wavelength - cp1e_ch17_p8)

7. $f = 7.04\times10^6\ \text{Hz} \because T = \frac{1}{f}$
(Chapter 16, Section (2) Period and Frequency in Oscillations - cp1e_ch16_ex3)

8. $a = 3.93\% \because Z = \rho v,\ a = \frac{(Z_2-Z_1)^2}{(Z_1+Z_2)^2}$
(Chapter 17, Section (7) Ultrasound - cp1e_ch17_ex7)

9. $d = 5.90\ \text{m} \because \Delta E_t = W_{nc} = -f_k d,\ f_k = \mu_k N = \mu_k mg,\ PE = \frac{1}{2}kx^2$
(Chapter 16, Section (7) Damped Harmonic Motion - cp1e_ch16_ex7)

10. $f = 743.2\ \text{Hz} \because f_{obs} = f_s\left(\frac{v_w}{v_w \pm v_s}\right)$ (review the Doppler effect)
(Chapter 17, Section (4) Doppler Effect and Sonic Booms - cp1e_ch17_ex4)

11. $L = 0.3171\ \text{m} \because f_n = n\frac{v_w}{4L},\ v_w = \left(331\,\frac{\text{m}}{\text{s}}\right)\sqrt{\frac{T}{273\ \text{K}}}$
(Chapter 17, Section (5) Sound Interference and Resonance: Standing Waves in Air Columns - cp1e_ch17_ex5a)

12. $v_{max} = 0.791\,\frac{\text{m}}{\text{s}} \because v(t) = \sqrt{\frac{k}{m}}X\sin\frac{2\pi t}{T}$ and the max of sine's range is 1.
(Chapter 16, Section (5) Energy and the Simple Harmonic Oscilltator - cp1e_ch16_ex6)